

CS401: Deep Learning — Transformers, Attention, and RAG Architectures

1. The Transformer Architecture and Attention Mechanism

Introduced in the seminal paper 'Attention Is All You Need' by Vaswani et al. (2017), the Transformer architecture replaced recurrent neural networks (RNNs) and LSTMs with attention mechanisms, enabling parallel training over long sequences. The core computational module is Scaled Dot-Product Attention, defined mathematically as: $\text{Attention}(Q, K, V) = \text{softmax}((Q * K^T) / \sqrt{d_k}) * V$, where Q denotes the Queries matrix, K denotes the Keys matrix, V denotes the Values matrix, and d_k is the dimension of the keys.

2. Multi-Head Attention and Positional Encodings

Multi-Head Attention projects Queries, Keys, and Values into h distinct linear subspaces, computes attention in parallel across each head, concatenates the outputs, and projects through a final linear layer. This allows the model to simultaneously attend to information from different representation subspaces at different positions. Because attention is permutation-invariant, Transformers inject Positional Encodings (sinusoidal functions or learnable embeddings like RoPE - Rotary Position Embedding) into input embeddings to preserve token order information.

3. Encoder-Decoder vs. Decoder-Only Models

Transformers exist in three main configurations: 1) Encoder-Only models (e.g., BERT) use bidirectional self-attention to generate rich representations for classification and embeddings; 2) Encoder-Decoder models (e.g., T5, BART) process an input sequence with an encoder and autoregressively generate output sequences with a decoder, suitable for translation; 3) Decoder-Only models (e.g., GPT, Llama, Mistral) utilize causal masked self-attention where tokens can only attend to previous tokens, making them the standard architecture for modern Large Language Models (LLMs).

4. Retrieval-Augmented Generation (RAG) Systems

Retrieval-Augmented Generation (RAG) is a hybrid AI architecture that combines dense vector retrieval with autoregressive LLM generation. Instead of relying solely on the parametric knowledge encoded in model weights, RAG dynamically retrieves relevant external context from vector databases (such as ChromaDB) using semantic embeddings (such as Sentence Transformers). The retrieved context is inserted into the LLM system prompt, grounding the generation in verifiable facts, drastically reducing hallucinations, and enabling citations to original source documents.